

Digital Lock-In Amplifier

PCB Project — Single Page Overview

Muthyala Koushik — EE24BTECH11044
Geedi Harshavardhan — EE24BTECH11025
Aslin Garvasis — EE24BTECH11008

DELIVERABLES

Hardware: A fully assembled 86 x 74mm Digital Lock-In Amplifier PCB using the STM32H743 microcontroller for real-time digital signal processing

Key Features:

- BNC-based signal and reference inputs
- I/Q synchronous detection (10 Hz - 50 kHz)
- Microvolt-level signal detection capability
- USB-powered operation

Validation:

- Schematic review and simulation-based analysis
- Hardware testing planned post-fabrication using signal generator and oscilloscope

SYSTEM SPECIFICATIONS

Frequency Range	10 Hz – 50 kHz
ADC Resolution	12-bit
Sampling Rate	Up to 1 MSPS
Gain Range	×10 to ×1000
Signal Detection	Microvolt-level
Phase Measurement	I/Q Method

ICs / COMPONENTS REQUIRED

● STM32H743ZIT6	Microcontroller — DSP + ADC core
● INA826	Instrumentation Amplifier (signal front-end)
● OPA350 / OPA2350	Op-Amp — filtering & buffering
● AMS1117 / TLV70033	3.3V Voltage Regulator
● BAT54 Diodes	Input Protection
● Ferrite Bead	Power Line Noise Filtering
● Micro-USB Connector	Power + Data Communication

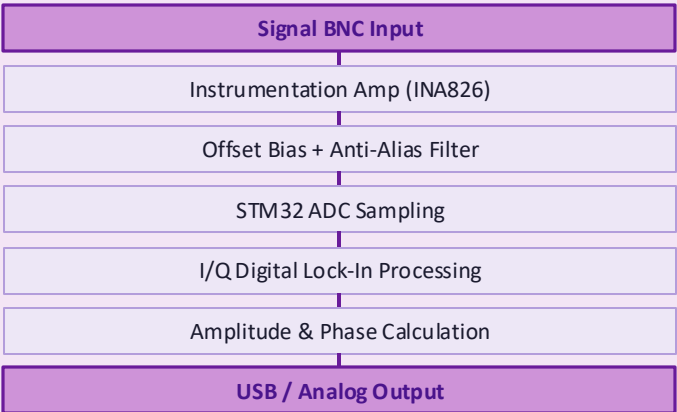
PROJECT COST BREAKDOWN

Service	Amount
PCB Fabrication (4-layer)	₹ 4,976
Components / BOM	₹ 14,336
PCB Assembly	₹ 2,630
Misc / Other Components	approx. ₹ 2,000

ESTIMATED TOTAL: ₹ 23,000 – 24,000

*Fabrication: 7WD lead time | Tax: 18% included

SIGNAL FLOW — BLOCK DIAGRAM



DRAFT PCB LAYOUT

